Teacher Guide — senior / module-14-magic-squares

IKS Curriculum

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Overview

Magic Squares — Bhadragaṇita

Band: Senior (Grades 9–12) · **Sessions:** 10 × 45 min · **Sanskrit:** Bhadra-gaṇita

Overview

A magic square is an n×n grid of distinct integers where every row, column, and main diagonal sums to the same constant. Nārāyaṇa Paṇḍita's *Gaṇita-kaumudī* (1356 CE), Book 14 (*Bhadra-gaṇita*), gives the most systematic pre-modern Indian treatment, including the stair-step (kothumi) method for odd-order squares. Cross-cultural comparison with Lo Shu (China) and Dürer (Europe) shows the universality of the puzzle and the Indian contribution to its mathematical systematisation.

NEP 2020 alignment

- §4.6 — Integration of Indian Knowledge Systems
- §4.23 — Experiential learning (constructive puzzling)
- §11.8 — Holistic, multidisciplinary (math + history + cross-cultural)

Key terms (this module)

- *bhadra-gaṇita* — "auspicious arithmetic" — the Indian term for magic squares
- *siddhāṅka (modern coinage)* — the common sum of all rows/columns/diagonals
- *kothumi vidhi* — Nārāyaṇa's algorithm for odd-order squares

Prerequisites

Module 2 (Panchabhūta) is recommended before this module. Modules 5 (Bindu Paddhati / dot addition) and 7 (×11 Multiplication) are useful prerequisites for Module 9 and 11.

Materials needed (across 10 days)

- Notebook + pencils per student
- Whiteboard / chart paper

- (See `lesson-plan.md` for per-day material lists)

10-day map

- **Day 1** — Discover the 3×3: Build the unique 3×3 magic square with 1–9 by trial.
- **Day 2** — The Magic Constant: Derive why 3×3 with 1–9 must sum to 15 per row.
- **Day 3** — How Many 3×3 Magic Squares?: Count the equivalence classes (up to symmetry).
- **Day 4** — Stair-Step Method (Nārāyaṇa): Apply Nārāyaṇa's algorithm to build a 5×5 square.
- **Day 5** — Practice — 5×5 and 7×7: Use the method on multiple odd-order squares.
- **Day 6** — 4×4 and the Khajuraho Square: Examine the Khajuraho inscription and its panel-magic property.
- **Day 7** — Historical Context — Nārāyaṇa and Predecessors: Place *Gaṇita-kaumudī* in 14th c. Indian math.
- **Day 8** — Cross-Cultural — Lo Shu and Dürer: Compare Indian, Chinese (Lo Shu), and European (Dürer 1514) magic squares.
- **Day 9** — Design Your Own: Create a personalised 4×4 or 5×5 magic square as visual art.
- **Day 10** — Final Share and Assessment: Present design + summative quiz.

Files in this pack

- `lesson-plan.md` — detailed 10-day arc
- `sources.md` — sources cited in this module
- `teacher-notes.md` — common misconceptions, differentiation
- `parent-guide.md` — what parents should know
- `days/day-NN-*.md` — 10 per-day lesson plans
- `quizzes/formative.md`, `quizzes/summative.md`
- `activities/activity-01-*.md`
- `assessments/rubric.md`, `assessments/project-brief.md`
- `student-workbook.md`
- `writing-prompts.md` (or `oral-recall.md` for Primary)
- `homework.md`
- `slides/deck.md`

Voice

Analytical, comparative, primary-source aware; 20–30 words on average. Pitches the concept first, frames the tradition as historical context, and uses IAST for all Sanskrit terms.

10-Day Lesson Plan

Lesson Plan — Magic Squares — Bhadragaṇita (Senior)

10-day arc, 45 min per session.

Day 1 — Discover the 3×3

Objective: Build the unique 3×3 magic square with 1–9 by trial.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 2 — The Magic Constant

Objective: Derive why 3×3 with 1–9 must sum to 15 per row.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 3 — How Many 3×3 Magic Squares?

Objective: Count the equivalence classes (up to symmetry).

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 4 — Stair-Step Method (Nārāyaṇa)

Objective: Apply Nārāyaṇa's algorithm to build a 5×5 square.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 5 — Practice — 5×5 and 7×7

Objective: Use the method on multiple odd-order squares.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 6 — 4×4 and the Khajuraho Square

Objective: Examine the Khajuraho inscription and its panel-magic property.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 7 — Historical Context — Nārāyaṇa and Predecessors

Objective: Place *Gaṇita-kaumudī* in 14th c. Indian math.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 8 — Cross-Cultural — Lo Shu and Dürer

Objective: Compare Indian, Chinese (Lo Shu), and European (Dürer 1514) magic squares.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 9 — Design Your Own

Objective: Create a personalised 4×4 or 5×5 magic square as visual art.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 10 — Final Share and Assessment

Objective: Present design + summative quiz.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Pacing notes

- Senior sessions are 45 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see `assessments/project-brief.md`) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see `teacher-notes.md` for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Teacher Notes

Teacher Notes — Magic Squares — Bhadragaṇita (Senior)

Common student misconceptions

- *"This is religious, not scientific."* — Frame the tradition as historical observation. Pañcabhūta-like classifications of matter exist in many cultures. Indian astronomy and math are documented, peer-reviewed historical fields.
- *"The Sanskrit terms are just jargon."* — Insist on the triple format (Sanskrit / IAST / meaning) once, then use the Sanskrit alone. The terms are precise; the discipline of using them properly is itself a learning outcome.
- *"If it's traditional, it must work."* — Distinguish empirical claims (testable) from ethical/aesthetic frames (which need separate justification). Encourage students to ask "how would I test this?".

Differentiation strategies

- **Extension for advanced students:** offer one optional "deeper read" per day — usually a primary source or modern paper.

- **Support for students needing scaffolds:** more pair-work, allow oral responses, provide partial outlines for writing prompts.
- **Multilingual classrooms:** Sanskrit terms are universal; vernacular equivalents (Hindi, Tamil, Telugu, Bengali, Marathi etc.) should be welcomed.

Safety

- Any activity involving food, plants, fire, or outdoor work needs explicit safety briefing.
- See the relevant day file for activity-specific safety notes.

Honest framing

- Where modern science clearly extends or revises traditional claims, say so.
- Where the tradition's framing is best understood as ethical/aesthetic, present it as such (not as competing science).
- Where attribution is debated (e.g., the "Vedic" status of Tirthaji's mathematics), present the debate honestly.

End-of-module reflection (for teachers)

- Which concepts landed well? Which didn't?
- Which students grew most? Which need a different approach next module?
- Add notes to `teacher-notes.md` for the next teacher who uses this pack.

Sources

Sources — Magic Squares — Bhadragaṇita (Senior)

Primary and secondary sources

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī* (1356 CE), Book 14 (Bhadra-gaṇita)
- Kim Plofker, *Mathematics in India* (Princeton, 2009), ch. 5
- Khajuraho temple inscription (~10th–11th c. CE) — 4×4 panel-magic square
- NCERT Mathematics Class VII, ch. 2 (Fractions and Decimals) — related grid arithmetic

A note on citation discipline

Per the curriculum's style guide, citations must be verifiable. If a verse number, page, or attribution cannot be confirmed, the source is paraphrased without invented attribution. Where direct quotes appear in Senior-band day files, the source is identified in IAST + edition.

For deeper reading

- The shared glossary at `curriculum/_shared/glossary.md` contains the IAST + meaning for every Sanskrit term used.
- Other modules in this curriculum that connect: see the module README's "Prerequisites" section.

Day 1 — Discover the 3×3

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: build the unique 3×3 magic square with 1–9 by trial.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. ""auspicious arithmetic" — the Indian term for magic squares")

Primary source (today's reading)

From *Nārāyaṇa Paṇḍita* — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *"What's one thing you remember about yesterday's lesson?"*
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day's idea is — a number trick, a celestial pattern, a herb's identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see `activities/`). Today's slice: focus on the part of the activity that reinforces Discover the 3×3.

Wrap-up (5 min)

- One-sentence exit ticket: *"What's one thing you learned today?"*
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 2 — The Magic Constant

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: derive why 3×3 with 1–9 must sum to 15 per row.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *siddhāṅka (modern coinage)* (IAST: siddhāṅka (modern coinage); lit. "the common sum of all rows/columns/diagonals")

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *"What's one thing you remember about yesterday's lesson?"*
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day's idea is — a number trick, a celestial pattern, a herb's identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.

4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see `activities/`). Today's slice: focus on the part of the activity that reinforces The Magic Constant.

Wrap-up (5 min)

- One-sentence exit ticket: *"What's one thing you learned today?"*
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 3 — How Many 3×3 Magic Squares?

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: count the equivalence classes (up to symmetry).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *kothumi vidhi* (IAST: kothumi vidhi; lit. "Nārāyaṇa's algorithm for odd-order squares")

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *“What’s one thing you remember about yesterday’s lesson?”*
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces How Many 3×3 Magic Squares?.

Wrap-up (5 min)

- One-sentence exit ticket: *“What’s one thing you learned today?”*
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 4 — Stair-Step Method (Nārāyaṇa)

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: apply nārāyaṇa's algorithm to build a 5×5 square.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. ""auspicious arithmetic" — the Indian term for magic squares")

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *"What's one thing you remember about yesterday's lesson?"*
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day's idea is — a number trick, a celestial pattern, a herb's identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see `activities/`). Today's slice: focus on the part of the activity that reinforces Stair-Step Method (Nārāyaṇa).

Wrap-up (5 min)

- One-sentence exit ticket: *"What's one thing you learned today?"*
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.

- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 5 — Practice — 5×5 and 7×7

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: use the method on multiple odd-order squares.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *siddhāṅka (modern coinage)* (IAST: siddhāṅka (modern coinage); lit. "the common sum of all rows/columns/diagonals")

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *"What's one thing you remember about yesterday's lesson?"*
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day's idea is — a number trick, a celestial pattern, a herb's identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see `activities/`). Today's slice: focus on the part of the activity that reinforces Practice — 5×5 and 7×7.

Wrap-up (5 min)

- One-sentence exit ticket: *“What’s one thing you learned today?”*
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 6 — 4×4 and the Khajuraho Square

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: examine the khajuraho inscription and its panel-magic property.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *kothumi vidhi* (IAST: kothumi vidhi; lit. “Nārāyaṇa’s algorithm for odd-order squares”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *“What’s one thing you remember about yesterday’s lesson?”*
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day's idea is — a number trick, a celestial pattern, a herb's identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see `activities/`). Today's slice: focus on the part of the activity that reinforces 4×4 and the Khajuraho Square.

Wrap-up (5 min)

- One-sentence exit ticket: *"What's one thing you learned today?"*
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 7 — Historical Context — Nārāyaṇa and Predecessors

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: place *gaṇita-kaumudī* in 14th c. indian math.

Materials

- Whiteboard / chart paper
- Notebook, pen per student

- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. ““auspicious arithmetic” — the Indian term for magic squares”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *“What’s one thing you remember about yesterday’s lesson?”*
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Historical Context — Nārāyaṇa and Predecessors.

Wrap-up (5 min)

- One-sentence exit ticket: *“What’s one thing you learned today?”*
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 8 — Cross-Cultural — Lo Shu and Dürer

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compare indian, chinese (lo shu), and european (dürer 1514) magic squares.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *siddhāṅka (modern coinage)* (IAST: siddhāṅka (modern coinage); lit. "the common sum of all rows/columns/diagonals")

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *"What's one thing you remember about yesterday's lesson?"*
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day's idea is — a number trick, a celestial pattern, a herb's identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see `activities/`). Today's slice: focus on the part of the activity that reinforces Cross-Cultural — Lo Shu and Dürer.

Wrap-up (5 min)

- One-sentence exit ticket: *"What's one thing you learned today?"*
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 9 — Design Your Own

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: create a personalised 4×4 or 5×5 magic square as visual art.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *kothumi vidhi* (IAST: kothumi vidhi; lit. "Nārāyaṇa's algorithm for odd-order squares")

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *"What's one thing you remember about yesterday's lesson?"*
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day's idea is — a number trick, a celestial pattern, a herb's identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.

4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see `activities/`). Today's slice: focus on the part of the activity that reinforces Design Your Own.

Wrap-up (5 min)

- One-sentence exit ticket: *"What's one thing you learned today?"*
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 10 — Final Share and Assessment

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: present design + summative quiz.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. ""auspicious arithmetic" — the Indian term for magic squares")

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *“What’s one thing you remember about yesterday’s lesson?”*
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Final Share and Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: *“What’s one thing you learned today?”*
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Formative Quiz — Magic Squares — Bhadragaṇita (Senior)

When: mid-module (after Day 5). **Duration:** 25 minutes. **Format:** mix of short answer, multiple choice, and reflection.

Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.
2. State the central idea of Magic Squares — Bhadragaṇita in one sentence.
3. (Multiple choice) Which day in this module focused on 'How Many 3×3 Magic Squares?'? (a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4
4. True / False — and explain in one sentence: *Magic Squares — Bhadragaṇita is purely a religious concept and has no scientific framing.*
5. Connect Magic Squares — Bhadragaṇita to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish 'bhadra-gaṇita' from related concepts? Give one example.
8. (Short answer) In your own words, explain Day 3's concept.
9. Identify **one** common misconception about Magic Squares — Bhadragaṇita and rebut it in one sentence.
10. Write **one** question you have about this module so far — something you want clarified before the summative.

How this is used

This quiz is **formative** — it's a check-in to see what's landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

Summative Quiz — Magic Squares — Bhadragaṇita (Senior)

When: end of module (after Day 10). **Duration:** 45 minutes. **Total points:** 30.

Questions

Part A — Vocabulary (~5 points) A1. Define *bhadra-gaṇita* in your own words. A2. Define *siddhāṅka (modern coinage)* in your own words. A3. Define *kothumi vidhi* in your own words.

Part B — Concepts (~10 points) B1. Explain the central idea of Magic Squares — Bhadragaṇita in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain why it matters. B4. Compare Magic Squares — Bhadragaṇita with a concept from another module (your choice). What's similar, what's different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

Part C — Application (~5 points) C1. Give an example of Magic Squares — Bhadragaṇita from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

Part D — Reflection (~5 points; ungraded for content) D1. What surprised you most in this module? D2. What's one question you still have?

Part E — Primary source (~5 points; Senior only) E1. Quote or paraphrase one passage from a primary source cited in this module. Explain it in your own words.

Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.
- Reflection: ungraded for content; awarded for thoughtfulness.

Pass mark

A score of 18/30 indicates the student has met the module's learning objectives.

Homework

Homework — Magic Squares — Bhadragaṇita (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — Discover the 3×3** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — The Magic Constant** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — How Many 3×3 Magic Squares?** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Stair-Step Method (Nārāyaṇa)** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Practice — 5×5 and 7×7** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — 4×4 and the Khajuraho Square** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Historical Context — Nārāyaṇa and Predecessors** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Cross-Cultural — Lo Shu and Dürer** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Design Your Own** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Share and Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.

- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

Writing Prompts

Writing Prompts — Magic Squares — Bhadragaṇita (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Magic Squares — Bhadragaṇita to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Magic Squares — Bhadragaṇita with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Magic Squares — Bhadragaṇita? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are doṣas a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Activity 1 — Class 3×3 Magic-Square Tournament

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Duration:** 25 min

Purpose

Reinforce the day-by-day concepts of Magic Squares — Bhadragaṇita through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

grids on paper, pencils, timer

What students do

In pairs, students race to construct the 3×3 magic square with 1–9 from a starting 9-position permutation. First pair with correct sums wins.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (15 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Magic Square as Visual Art

Module: Magic Squares — Bhadragaṇita · **Band:** Senior

What you will do

Design a 4×4 or 5×5 magic square. Decorate it as a piece of visual art (mandala-style, calligraphic, geometric). Include the magic constant calculation on the back.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (500 words (Senior)): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (5-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 3 sources. Use the form: *Author*, *Title*, *year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

Rubric

Assessment Rubric — Magic Squares — Bhadragaṇita (Senior)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

Criteria

Accuracy of concepts — uses Sanskrit terms correctly, gets the central idea right

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Source discipline — cites sources, distinguishes traditional framing from empirical claim

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Application — connects the module to other modules and to real-world examples

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Presentation — clear, organised, evidence-supported

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Reflection — honest, thoughtful, identifies what was learned and what remains uncertain

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.
3. Share at least one strength and one area for growth with each student.

Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher’s assessment. The teacher and student then meet to compare.